

# CG Infinity - Git & GitHub Hands-on Assignment

**Project:** Employee Management System

**Submitted By:** Sahil

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## Task 1

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**Objective:** Create a new project folder named EmployeeManagementSystem and initialize it as a Git repository.

```
C:\Users\SAHIL cd EmployeeManagementSystem
C:\Users\SAHIL EmployeeManagement System>git init
Reinitialized existing Git repository in C:/Users/SAHIL/Employee
ManagementSystem/.git/
C:\Users\SAHIL Employee ManagementSystem>git status
On branch master
No commits yet
nothing to commit (create/copy files and use "git add" to track)
```

## Task 2

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**Objective:** Create README.md, index.html and style.css. Add project title and description in README.md.

```
C:\Users\SAHIL Employee Management System>type nul >README.md
C:\Users\SAHIL EmployeeManagementSystem>type nul index.html
C:\Users\SAHIL EmployeeManagementSystem>type nul >style.css
C:\Users\SAHIL Employee ManagementSystem>dir
Volume in drive C is Windows
Volume Serial Number is DC68-721D
Directory of C:\Users\SAHIL EmployeeManagement System

22-06-2026 00:23 <DIR>
22-06-2026 00:04 <DIR>
22-06-2026 00:23 index.html
22-06-2026 00:23 Readme.md
22-06-2026 00:23 style.css
3 File(s) 0 bytes
2 Dir(s) 33,242,787,840 bytes free
C:\Users\SAHIL EmployeeManagementSystem>git status
On branch master
```

## Task 3

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**Objective:** Verify repository status and identify untracked files.

```
No commits yet
Untracked files:
(use "git add <file>..." to include in what will be committed)
Readme.md
index.html
style.css
nothing added to commit but untracked files present (use "git add" to track)
```

## Task 4

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**Objective:** Add files to staging area and create the first commit.

```
C:\Users\SAHIL EmployeeManagement System>git add
C:\Users\SAHIL EmployeeManagement System>git status
On branch master
No commits yet
Changes to be committed:
(use "git rm --cached <file>..." to unstage)
new File: Readme.md
new File: index.html
new File: style.css

C:\Users\SAHIL Employee ManagementSystem>git commit -m "Initial project setup"
[master (root-commit) ca17ed3] Initial project setup
3 files changed, 6 insertions(+)
create mode 100644 Readme.md
create mode 100644 index.html
create mode 100644 style.css
```

## Task 5

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**Objective:** Review repository history using git log.

```
C:\Users\SAHIL Employee Management System>git log
commit ca17ed36077ec4145f42b9c26821008d5aa6de21 (HEAD -> master)
Author: SAHIL58 <sahil89408@gmail.com>
Date: Mon Jun 22 00:28:59 2026 +0530
Initial project setup
C:\Users\SAHIL Employee ManagementSystem>
```

## Task 6

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**Objective:** Create feature/employee-registration branch and switch to it.

```
C:\Users\SAHIL EmployeeManagementSystem>git branch featue/employee-registration
C:\Users\SAHIL Employee ManagementSystem>git switch featue/employee_registration
fatal: invalid reference: featue/employee_registration
C:\Users\SAHIL Employee ManagementSystem>git switch featue/employee-registration
Switched to branch 'featue/employee-registration'
C:\Users\SAHIL EmployeeManagementSystem>
```

## Task 7

**Objective: Create employee-registration.html and commit changes.**

```
C:\Users\SAHIL EmployeeManagementSystem>notepad employee-registration.html
C:\Users\SAHIL EmployeeManagementSystem>git status
On branch featue/employee-registration
Untracked files:
(use "git add <file>..." to include in what will be committed)
employee-registration.html
nothing added to commit but untracked files present (use "git add" to track)
C:\Users\SAHIL EmployeeManagementSystem>git commit -m "Added employee registration page"
On branch featue/employee-registration
Untracked files:
(use "git add <file>..." to include in what will be committed)
employee-registration.html
nothing added to commit but untracked files present (use "git add" to track)
C:\Users\SAHIL EmployeeManagementSystem>git add employee-registration.html
C:\Users\SAHIL Employee ManagementSystem>git commit -m "Added employee registration page"
[featue/employee-registration 2881973] Added employee registration page
1 file changed, 9 insertions(+)
create mode 100644 employee-registration.html
```

## Task 8

**Objective: Compare working directory changes with last committed version using git diff.**

```
C:\Users\SAHIL Employee ManagementSystem>notepad employee-registration.html
C:\Users\SAHIL EmployeeManagementSystem>git diff
diff --git a/employee-registration.html b/employee-registration.html
index ffd5033..c02345c 100644
--- a/employee-registration.html
+++ b/employee-registration.html
@@ -5,5 +5,6 @@
</head>
<body>
<h1>Employee Registration Form</h1>
+<h2>Register New Employee</h2>
</body>
</html>
\ No newline at end of file
```

## Task 9

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**Objective:** Merge feature/employee-registration branch into main/master.

```
C:\Users\SAHIL EmployeeManagementSystem>git switch master
Switched to branch 'master'
C:\Users\SAHIL Employee Management System>git merge featue?employee-registration
merge: featue?employee-registration not something we can merge
C:\Users\SAHIL EmployeeManagement System>git merge featue/employee-registration
Updating cal7ed3..59a0388
Fast-forward
 employee-registration.html | 10 +
1 file changed, 10 insertions(+)
create mode 100644 employee-registration.html
```

## Task 10

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**Objective:** Create GitHub repository and connect local repository.

```
C:\Users\SAHIL EmployeeManagementSystem>git remote add origin https://github.com/
SAHIL58/EmployeeManagementSystem
C:\Users\SAHIL Employee Management System>git remote -v
origin https://github.com/SAHIL58/EmployeeManagementSystem (fetch)
origin https://github.com/SAHIL58/EmployeeManagementSystem (push)
C:\Users\SAHIL EmployeeManagement System>
```

## Task 11

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**Objective:** Publish project to GitHub.

```
C:\Users\SAHIL Employee ManagementSystem>git push u origin master
Enumerating objects: 10, done.
Counting objects: 100% (10/10), done.
Delta compression using up to 16 threads
Compressing objects: 100% (9/9), done.
Writing objects: 100% (10/10), 973 bytes | 324.00 KiB/s, done.
Total 10 (delta 3), reused (delta 0), pack-reused
remote: Resolving deltas: 100% (3/3), done.
To (from 0) https://github.com/SAHIL58/EmployeeManagementSystem
* [new branch] master -> master
branch 'master' set up to track 'origin/master'.
```

## Task 12

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**Objective:** Create feature/employee-dashboard branch and commit dashboard page.

```

C:\Users\SAHIL Employee ManagementSystem>git branch feature/employee-dashboard
C:\Users\SAHIL Employee ManagementSystem>git switch feature/employee-dashboard
Switched to branch feature/employee-dashboard
C:\Users\SAHIL Employee ManagementSystem>type nul employee-dashboard.html
C:\Users\SAHIL EmployeeManagement System>notepad employee-dashboard.html
C:\Users\SAHIL Employee ManagementSystem>git add.
C:\Users\SAHIL Employee ManagementSystem>git commit -m"Added employee dashboard page"
[feature/employee-dashboard bffb2a3] Added employee dashboard page
1 file changed, 9 insertions(+)
create mode 100644 employee-dashboard.html

```

## Task 13

**Objective: Synchronize local repository with latest GitHub changes.**

```

C:\Users\SAHIL Employee ManagementSystem>git switch master
Switched to branch 'master'
Your branch is up to date with 'origin/master'.
C:\Users\SAHIL Employee Management System>git pull origin master
From https://github.com/SAHIL58/EmployeeManagementSystem
* branch
Already up to date.
master -> FETCH_HEAD

```

## Task 14

**Objective: Merge dashboard feature into main/master branch.**

```

C:\Users\SAHIL Employee Management System>git merge feature/employee-dashboard
Updating 59a0388..bffb2a3
Fast-forward
 employee-dashboard.html | 9
1 file changed, 9 insertions(+)
create mode 180644 employee-dashboard.html
C:\Users\SAHIL EmployeeManagementSystem>dir
Volume in drive C is Windows
Volume Serial Number is DC68-7210
Directory of C:\Users\SAHIL Employee Management System

22-06-2026 00:52 <DIR>
22-06-2026 00:51 <DIR>
22-06-2026 00:52 138 employee-dashboard.html
22-06-2026 00:44 181 employee-registration.html
22-06-2026 00:23 index.html
22-06-2026 00:26 117 Readme.md
22-06-2026 00:23 style.css
5 File(s) 436 bytes
2 Dir(s) 33,222,971,392 bytes free

```

## Task 15

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**Objective:** Display all branches and identify active branch.

```
C:\Users\SAHIL Employee ManagementSystem>git branch
  featue/employee-registration
  feature/employee-dashboard
* master
```

## Task 16

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**Objective:** Create and resolve merge conflict between login and dashboard branches.

```
C:\Users\SAHIL EmployeeManagement System>git switch feature/login
M Readme.md
Switched to branch 'feature/login'
C:\Users\SAHIL Employee ManagementSystem>notepad README.md
C:\Users\SAHIL EmployeeManagementSystem>git add README.md
C:\Users\SAHIL EmployeeManagement System>git commit -m "Updated README for login
feature"
On branch feature/login
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
    Readme.md
no changes added to commit (use "git add" and/or "git commit -a")
C:\Users\SAHIL Employee ManagementSystem>git switch master
M Readme.md
Switched to branch 'master'
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)
C:\Users\SAHIL EmployeeManagement System>git switch feature employee-dashboard
fatal: only one reference expected
C:\Users\SAHIL Employee Management System>git switch feature/employee-dashboard
M Readme.md
Switched to branch 'feature/employee-dashboard'
C:\Users\SAHIL EmployeeManagement System>notepad README.md
C:\Users\SAHIL EmployeeManagementSystem>notepad README.md
C:\Users\SAHIL Employee Management System>git add README.md
C:\Users\SAHIL Employee Management System>git commit -m "Updated README for dashboard
feature"
On branch feature/employee-dashboard
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
C:\Users\SAHIL EmployeeManagementSystem>git add
C:\Users\SAHIL Employee Management System>git commit -m "Login feature update"
[login 962288b] Login feature update
 1 file changed, 1 insertion(+), 1 deletion(-)
C:\Users\SAHIL Employee ManagementSystem>git switch master
Switched to branch 'master'
Your branch is ahead of origin/master by 1 commit.
  (use "git push" to publish your local commits)
C:\Users\SAHIL EmployeeManagementSystem>git branch dashboard
C:\Users\SAHIL Employee ManagementSystem>git switch dashboard
Switched to branch 'dashboard'
C:\Users\SAHIL EmployeeManagementSystem>notepad README.md
C:\Users\SAHIL EmployeeManagementSystem>git add
C:\Users\SAHIL Employee ManagementSystem>git commit -m "Dashboard feature update"
[dashboard 918ef1a] Dashboard feature update
 1 file changed, 1 insertion(+), 1 deletion(-)
C:\Users\SAHIL EmployeeManagement System>git switch master
Switched to branch 'master'
Your branch is ahead of 'origin/master' by 1 commit.
  (use "git push" to publish your local commits)
C:\Users\SAHIL EmployeeManagement System>git merge Login
Updating bffb2a3..962288b
```

```

Fast-Forward
  Readme.md2
  1 file changed, 1 insertion(+), 1 deletion(-)
C:\Users\SAHIL EmployeeManagement System>git merge dashboard
Auto-merging Readme.md
CONFLICT (content): Merge conflict in Readme.md
Automatic merge failed; fix conflicts and then commit the result.
C:\Users\SAHIL Employee ManagementSystem>
Headm
tent)
gefallit Viww
SHAR
SHAR
Employee Management systes Login & Dashboard Features
SHARE staple b-based Employee Management Systée for managing employee records.
end
ting
opriat
obec
SHAR
ter
sahe
ush
arged
cts a
erge
dd f
ded t
SHAR
C:\Users\SAHIL EmployeeManagementSystem>git add README.md
C:\Users\SAHIL EmployeeManagement System>git status
De branch master
Your branch is ahead of origin/master by 2 commits.
  (use "git push to publish your local commits)
You have unmerged paths.
  (Fix conflicts and run "git commit )
  (use "git merge abort to aburt the serge)
Unserged paths:
  (use "git add <file>... to mark resolution)
no changes added to commit (use "git add" and/or "git commit -a )
C:\Users\SAHIL EmployeeManagementSystem>git add Readme.md
C:\Users\SAHIL EmployeeManagementSystem>git status
On branch master
Your branch is ahead of origin/master by 2 commits.
  (use "git push to publish your local commits)
All conflicts Fixed but you are still merging
  (use "git commit to conclude merge)
Changes to be committed:
C:\Users\SAHIL EmployeeManagement Systas
C:\Users\SAHIL EmployeeManagement System>git commit -m "Resolved merge conflict"
[master 78adid6) Resolved merge conflict
C:\Users\SAHIL EmployeeManagementSystem>git diff
C:\Users\SAHIL EmployeeManagementSystem>

```